

CE392-Project

Sliding-window CNN accelerator

Midterm Presentation

Presenter:

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Agenda

- Model Design & Presion Selection
- Hardware Architecture
- Verification
- TODO?
- References

Model Structure - MobileFaceNet

Input Image Size $3 \times 112 \times 112$

Input	Operator	t	c	n	s
$112^2 \times 3$	conv3x3	-	64	1	2
$56^2 \times 64$	depthwise conv3x3	-	64	1	1
$56^2 \times 64$	bottleneck	2	64	5	2
$28^2 \times 64$	bottleneck	4	128	1	2
$14^2 \times 128$	bottleneck	2	128	6	1
$14^2 \times 128$	bottleneck	4	128	1	2
$7^2 \times 128$	bottleneck	2	128	2	1
$7^2 \times 128$	conv1x1	-	512	1	1
$7^2 \times 512$	linear GDConv7x7	-	512	1	1
$1^2 \times 512$	linear conv1x1	-	128	1	1

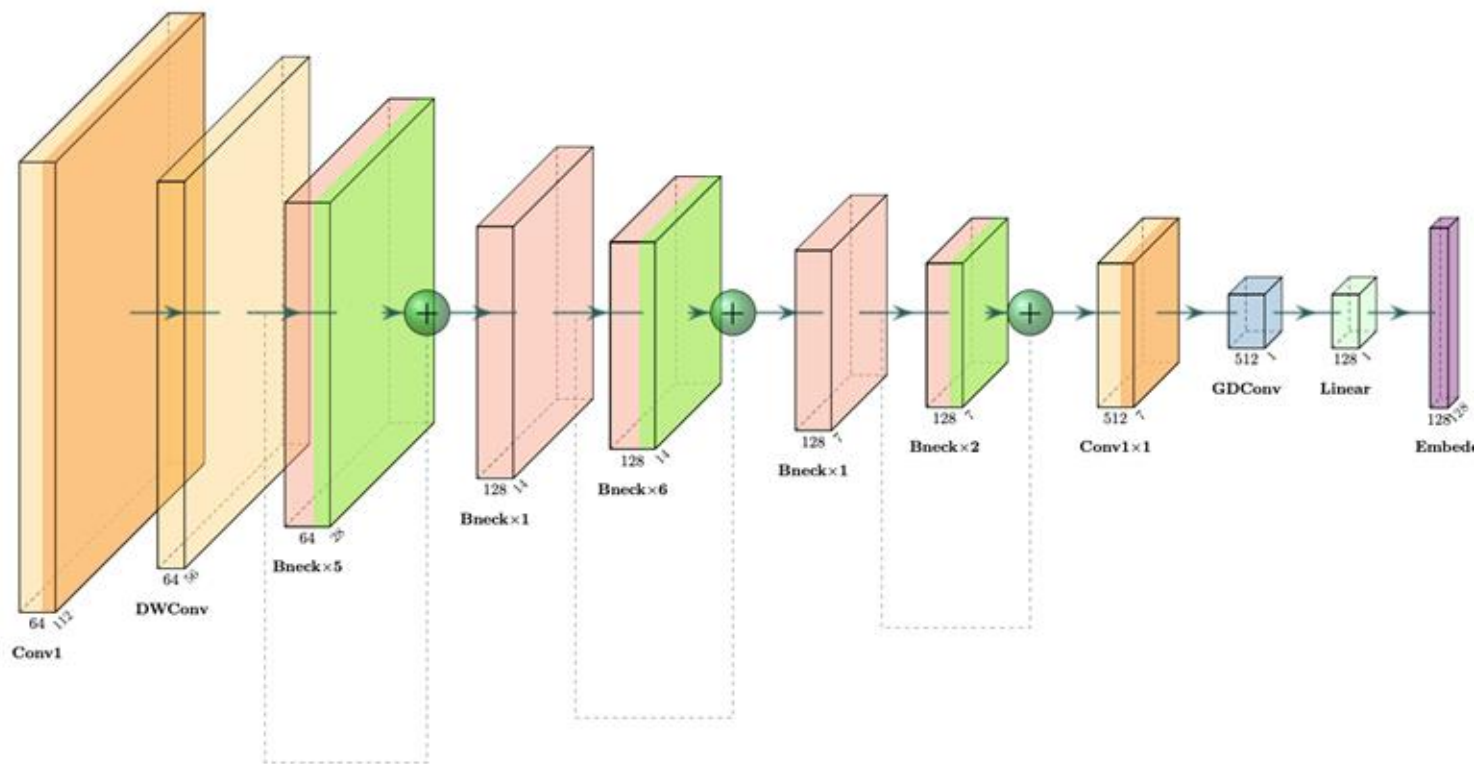
Input Image Size $3 \times 112 \times 96$

Input	Operator	t	c	n	s
$3 \times 112 \times 96$	Conv3 $\times$ 3	-	64	1	2
$64 \times 56 \times 48$	Depthwise Conv3 $\times$ 3 [14]	-	64	1	1
$64 \times 56 \times 48$	Bottleneck [14]	2	64	5	2
$64 \times 28 \times 24$	Bottleneck	4	128	1	2
$128 \times 14 \times 12$	Bottleneck	2	128	6	1
$128 \times 14 \times 12$	Bottleneck	4	128	1	2
$128 \times 7 \times 6$	Bottleneck	2	128	2	1
$128 \times 7 \times 6$	Conv1 $\times$ 1	-	512	1	1
$512 \times 7 \times 6$	GDConv7 $\times$ 6 [13]	-	512	1	1
$512 \times 1 \times 1$	Linear	-	128	1	1

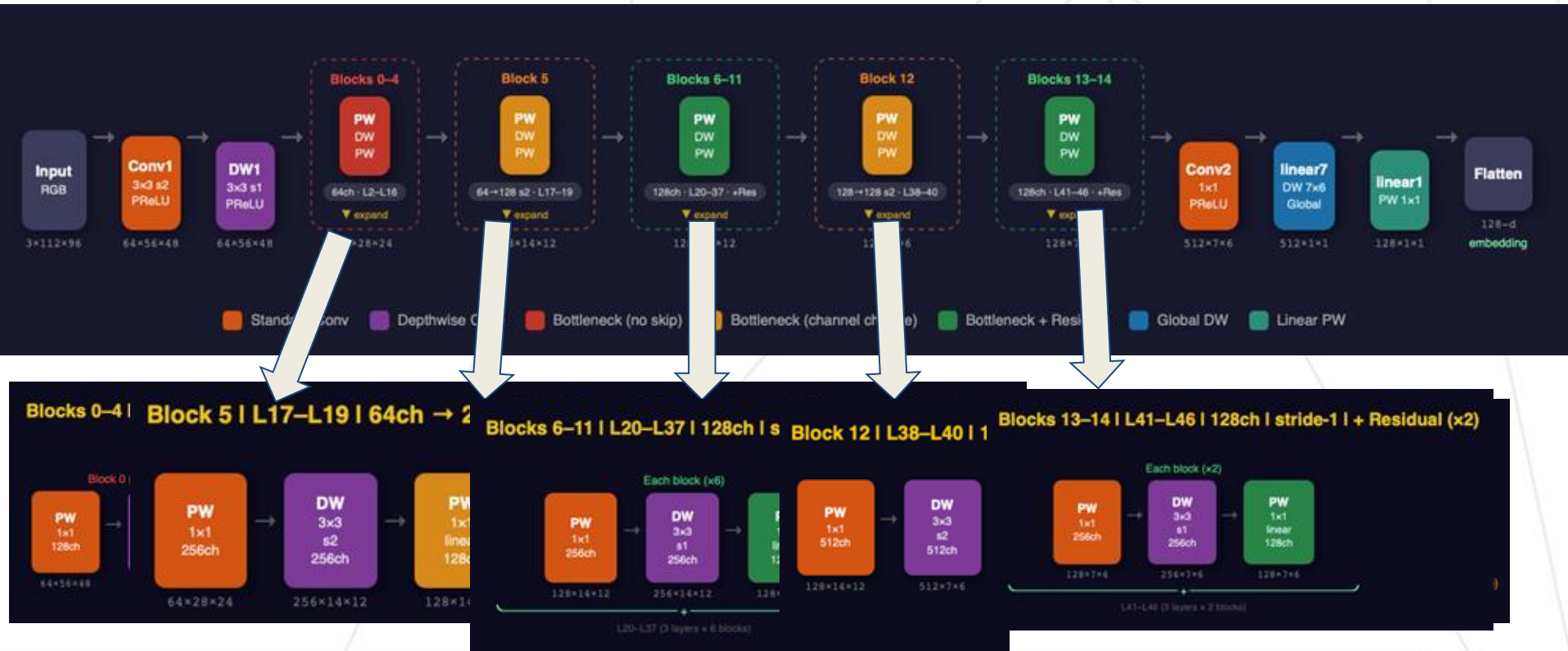
[Gradient-Enhanced Softmax for Face Recognition](#)

[MobileFaceNets: Efficient CNNs for Accurate Real-Time Face Verification on Mobile Devices](#)

Model Structure



Model Structure

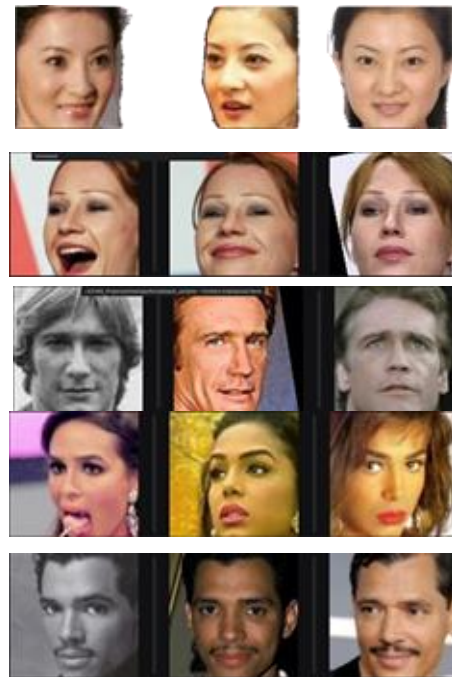


Precision Selection

- PyTorch and C float are almost identical, so the float implementation is consistent with the reference.
- C INT16** stays very close to PyTorch, with only small similarity differences across most pairs.
- C INT8 shows noticeably larger deviations, especially on several positive and negative pairs.
- The INT8 error is less stable, so it introduces more precision loss than INT16.
- We choose INT16 as the final design because it gives the balance between accuracy and precision.

Type	Indices	PyTorch	C Float	C INT16	C INT8	vs pyto	vs pyto	vs pyto
Positive	[381, 480]	0.8242	0.8242	0.8258	0.7838	0.000001 (OK)	0.001532 (OK)	0.040420 (OK)
Positive	[460, 739]	0.5098	0.5098	0.5080	0.4557	0.000004 (OK)	0.001805 (OK)	0.054116 (WARN)
Positive	[275, 765]	0.4646	0.4646	0.4623	0.4252	0.000001 (OK)	0.002298 (OK)	0.039325 (OK)
Positive	[320, 809]	0.5495	0.5495	0.5472	0.3068	0.000000 (OK)	0.002337 (OK)	0.242779 (WARN)
Positive	[419, 935]	0.6222	0.6222	0.6126	0.6703	0.000001 (OK)	0.009552 (OK)	0.048130 (OK)
Positive	[506, 983]	0.5071	0.5071	0.5027	0.5077	0.000006 (OK)	0.004427 (OK)	0.000581 (OK)
Positive	[425, 997]	0.5546	0.5546	0.5538	0.5486	0.000003 (OK)	0.000858 (OK)	0.005995 (OK)
Positive	[191, 1018]	0.6713	0.6713	0.6703	0.5151	0.000003 (OK)	0.001019 (OK)	0.156202 (WARN)
Positive	[329, 1029]	0.6433	0.6433	0.6440	0.5137	0.000001 (OK)	0.000762 (OK)	0.129544 (WARN)
Positive	[404, 1104]	0.6528	0.6528	0.6557	0.5532	0.000008 (OK)	0.002945 (OK)	0.099630 (WARN)
Negative	[381, 739]	-0.0156	-0.0156	-0.0089	0.1225	0.000006 (OK)	0.006698 (OK)	0.138106 (WARN)
Negative	[460, 765]	0.1402	0.1402	0.1407	0.3556	0.000001 (OK)	0.000574 (OK)	0.215400 (WARN)
Negative	[275, 809]	0.2342	0.2342	0.2308	0.3313	0.000000 (OK)	0.003388 (OK)	0.097039 (WARN)
Negative	[320, 935]	0.4617	0.4617	0.4611	0.3269	0.000004 (OK)	0.000613 (OK)	0.134818 (WARN)
Negative	[419, 983]	0.1141	0.1141	0.1175	0.1704	0.000006 (OK)	0.003461 (OK)	0.056324 (WARN)
Negative	[506, 997]	0.0821	0.0821	0.0824	0.2854	0.000006 (OK)	0.000259 (OK)	0.203290 (WARN)
Negative	[425, 1018]	0.2000	0.2000	0.2028	0.3035	0.000000 (OK)	0.002852 (OK)	0.103499 (WARN)
Negative	[191, 1029]	0.0977	0.0977	0.1008	0.4384	0.000002 (OK)	0.003102 (OK)	0.340676 (WARN)
Negative	[329, 1104]	0.2719	0.2719	0.2749	0.3667	0.000002 (OK)	0.002955 (OK)	0.094775 (WARN)
Negative	[404, 480]	0.0692	0.0692	0.0695	0.1273	0.000001 (OK)	0.000317 (OK)	0.058101 (WARN)

Model Eval - Original Dataset



C INT16 - Cosine Similarity Matrix

	person_10056_0	person_10056_1	person_10056_2	person_31753_0	person_31753_1	person_31753_2	person_37870_0	person_37870_1	person_37870_2	person_44334_0	person_44334_1	person_44334_2	person_54087_0	person_54087_1	person_54087_2	person_81370_0	person_81370_1	person_81370_2
person_10056_0	1.0000	0.6576	0.7161	-0.0562	-0.0158	0.0873	0.0453	0.1117	-0.0496	0.1036	0.0881	0.1111	0.0257	-0.0375	-0.0485	0.0799	-0.0748	0.1536
person_10056_1	0.6576	1.0000	0.6437	0.0526	0.0228	-0.0766	-0.0307	-0.0092	-0.0654	-0.0431	0.1722	0.0260	-0.1272	-0.0328	-0.0427	0.1130	-0.0291	0.1211
person_10056_2	0.7161	0.6437	1.0000	0.0264	0.1103	0.1201	0.0499	0.0853	-0.1140	0.0778	0.1272	0.1140	0.0045	0.0192	0.0240	0.1356	-0.0099	0.1541
person_31753_0	-0.0562	0.0526	0.0264	1.0000	0.7290	0.5476	0.1318	0.1426	0.1075	-0.0518	0.0327	0.0516	0.1357	0.1696	0.1422	0.0497	0.0390	-0.0208
person_31753_1	-0.0158	0.0228	0.1103	0.7290	1.0000	0.6899	0.1764	0.1974	0.1295	0.0884	0.2513	0.1499	0.1127	0.1359	0.2042	0.0844	-0.0166	0.0650
person_31753_2	0.0873	-0.0766	0.1201	0.5476	0.6899	1.0000	0.2515	0.1237	0.0274	-0.0061	0.0331	0.1044	0.0916	0.0269	0.0784	0.1290	-0.0909	0.1138
person_37870_0	0.0453	-0.0307	0.0499	0.1318	0.1764	0.2515	1.0000	0.6267	0.4034	0.0507	-0.0973	0.1494	-0.0130	-0.0206	-0.0998	0.3379	0.1920	0.2446
person_37870_1	0.1117	-0.0092	0.0853	0.1426	0.1974	0.1237	0.6267	1.0000	0.5165	-0.0040	-0.0417	0.1294	0.1116	0.0366	-0.0157	0.2536	0.0526	0.2346
person_37870_2	-0.0496	-0.0654	-0.1140	0.1075	0.1295	0.0274	0.4034	0.5165	1.0000	0.1113	0.1105	0.1142	-0.0830	-0.0905	-0.0191	0.2024	0.1111	0.1874
person_44334_0	0.1036	-0.0431	0.0778	-0.0518	0.0884	-0.0061	0.0507	-0.0040	0.1113	1.0000	0.3918	0.4026	0.1724	0.0771	-0.0123	0.0272	0.0701	0.0089
person_44334_1	0.0881	0.1722	0.1272	0.0327	0.2513	0.0331	-0.0973	-0.0417	0.1105	0.3918	1.0000	0.4189	0.1171	0.1462	0.1716	0.0474	0.0596	0.2054
person_44334_2	0.1111	0.0260	0.1140	0.0516	0.1499	0.1044	0.1494	0.1294	0.1142	0.4026	0.4189	1.0000	0.2443	0.2281	0.1413	0.1470	0.1450	0.1442
person_54087_0	0.0257	-0.1272	0.0045	0.1357	0.1127	0.0916	-0.0130	0.1116	-0.0030	0.1724	0.1171	0.2443	1.0000	0.6498	0.5477	-0.1169	-0.0904	-0.0245
person_54087_1	-0.0375	-0.0328	0.0192	0.1696	0.1359	0.0269	-0.0206	0.0366	-0.0905	0.0771	0.1462	0.2281	0.6498	1.0000	0.5876	0.0456	0.0592	-0.0263
person_54087_2	-0.0485	-0.0427	0.0240	0.1422	0.2042	0.0784	-0.0998	-0.0157	-0.0191	-0.0123	0.1716	0.1413	0.5477	0.5876	1.0000	-0.0346	0.1221	0.0508
person_81370_0	0.0799	-0.1130	0.1356	0.0497	0.0844	0.1290	0.3379	0.2536	0.2024	0.0272	0.0474	0.1470	-0.1169	0.0456	-0.0346	1.0000	0.4841	0.5438
person_81370_1	-0.0748	-0.0291	-0.0099	0.0390	-0.0166	-0.0909	0.1920	0.0526	0.1111	0.0701	0.0596	0.1450	-0.0904	0.0592	0.1221	0.4841	1.0000	0.4791
person_81370_2	0.1536	0.1211	0.1541	-0.0208	0.0650	0.1138	0.2446	0.2346	0.1874	0.0089	0.2054	0.1442	-0.0245	-0.0263	0.0508	0.5438	0.4791	1.0000

[C INT16] Per-person intra vs inter similarity

Person	Intra (same person)	Inter (diff person)
person_10056	avg 0.6725 min 0.6437	avg 0.0333 max 0.1722
person_31753	avg 0.6555 min 0.5476	avg 0.0796 max 0.2515
person_37870	avg 0.5155 min 0.4034	avg 0.0768 max 0.3379
person_44334	avg 0.4044 min 0.3918	avg 0.0892 max 0.2513
person_54087	avg 0.5950 min 0.5477	avg 0.0435 max 0.2443
person_81370	avg 0.5023 min 0.4791	avg 0.0812 max 0.3379
Overall intra avg : 0.5575		
Overall inter avg : 0.0673		
Separability gap : +0.4903 (6000 %)		

Model Eval - Self Prepared



C INT16 - Cosine Similarity Matrix													
	dan_1	dan_2	dan_3	elon_1	elon_2	elon_3	elon_4	trump_1	trump_2	trump_3	wil_1	wil_2	wil_3
dan_1	1.0000	0.4077	0.4135	-0.0782	0.1101	0.0007	0.1364	-0.0947	-0.1805	-0.0577	0.1279	0.2156	-0.0321
dan_2	0.4077	1.0000	0.6320	0.1266	0.1559	0.0875	0.1318	0.0201	0.0684	0.1434	0.3184	0.2680	0.1581
dan_3	0.4135	0.6320	1.0000	0.1889	0.1715	0.2244	0.1423	0.0199	0.1532	0.0669	0.0627	0.1541	0.1759
elon_1	-0.0782	0.1266	0.1889	1.0000	0.5675	0.4696	0.4843	0.1633	0.1646	0.2147	0.0470	0.0139	0.0431
elon_2	0.1101	0.1559	0.1715	0.5675	1.0000	0.5850	0.6821	0.1501	0.0859	0.2417	0.0601	0.1006	-0.0785
elon_3	0.0007	0.0875	0.2244	0.4696	0.5850	1.0000	0.5613	0.0464	0.1355	0.0576	-0.0035	-0.0051	-0.0468
elon_4	0.1364	0.1318	0.1423	0.4843	0.6821	0.5613	1.0000	0.1420	0.1473	0.1580	0.0530	0.1167	-0.0855
trump_1	-0.0947	0.0201	0.0199	0.1633	0.1501	0.0464	0.1420	1.0000	0.6259	0.4753	-0.0796	0.0753	-0.0107
trump_2	-0.1805	0.0684	0.1532	0.1646	0.0859	0.1355	0.1473	0.6259	1.0000	0.4952	-0.2009	0.0664	0.0427
trump_3	-0.0577	0.1434	0.0669	0.2147	0.2417	0.0576	0.1580	0.4753	0.4952	1.0000	0.0504	0.0953	-0.0132
wil_1	0.1279	0.3184	0.0627	0.0470	0.0601	-0.0035	0.0530	-0.0796	-0.2009	0.0504	1.0000	0.5494	0.4206
wil_2	0.2156	0.2680	0.1541	0.0139	0.1006	-0.0051	0.1167	0.0753	0.0664	0.0953	0.5494	1.0000	0.5358
wil_3	-0.0321	0.1581	0.1759	0.0431	-0.0785	-0.0468	-0.0855	-0.0107	0.0427	-0.0132	0.4206	0.5358	1.0000

[C INT16] Per-person intra vs inter similarity				
Person	Intra (same person)		Inter (diff person)	
dan	avg 0.4844	min 0.4077	avg 0.0995	max 0.3184
elon	avg 0.5583	min 0.4696	avg 0.0922	max 0.2417
trump	avg 0.5321	min 0.4753	avg 0.0624	max 0.2417
wil	avg 0.5019	min 0.4206	avg 0.0563	max 0.3184

Overall intra avg : 0.5270
Overall inter avg : 0.0783
Separability gap : +0.4487 (GOOD ✓)

Hardware Architecture

- A fixed-point, MobileFaceNet-specific convolution frontend accelerator with ROM-based layer configuration
- Operation types :
 - Standard Conv、Depthwise Conv、Pointwise Conv、Global DW Conv、PReLU、Residual Add

① DATAPATH

mfn_sliding_window

9-entry pixel buffer

mfn_mac_array

9 × 16-bit parallel MACs
40-bit dot product

mfn_activation

Q20 → Q10 · Clamp · PReLU
Residual Add

② CONTROL PATH

mfn_controller

Main FSM
Layer → Y,X → c_in → c_out

mfn_addr_gen

HWC address generation
Ping-pong SRAM offset

Layer Config ROM

50 × 64-bit layer configs
stride / ch / type / wgt_base

③ MEMORY SUBSYSTEM

Feature Map SRAM

1MB · 3-buffer ping-pong
HWC layout

Weight ROM

20-bit addr · ~1M int16
9 weights/cycle

Bias / PReLU ROM

14-bit addr · int32 bias
int16 Q10 alpha

Hardware - Controller

INNER LOOP PIPELINE

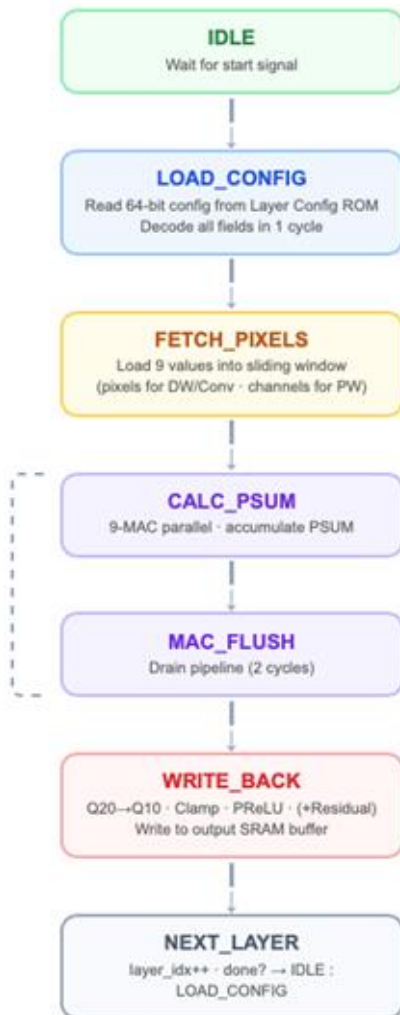


NESTED LOOP EXECUTION ORDER

```
for each layer (L0 → L49):
  for y in 0 → H_out-1:
    for x in 0 → W_out-1:
      for c_out in 0 → C_out-1:
        for c_in in 0 → C_in-1:
          fetch → MAC → accumulate → write
```

- layer — from Config ROM
- y, x — spatial position
- c_out — output channel
- c_in — input channel (innermost)

FSM STATE DIAGRAM



Hardware - address gen?

- HWC formula :

$$\text{Addr} = (y \cdot W + x) \cdot C + c$$

- $y = \text{row_idx}$ $x = \text{col_idx}$
- C/c: total channel/channel idx
- W: feature map width
 - $\text{Pixel_offset} = y \cdot W + x$
 - $\text{Addr} = \text{pixel_offset} \cdot C + c$

- Ping-Pong Extension

- Each buffer's base address is offset by 344,064 words
- Write address = HWC addr + ping-pong base offset

✗ CHW LAYOUT (CHANNEL-FIRST)

Memory order: all R pixels → all G pixels → all B pixels



△ To read R, G, B at position $(y=2, x=3)$: addresses are **far apart** ($H \times W$ apart each)
→ Stride = $H \times W$ per channel jump → complex address logic

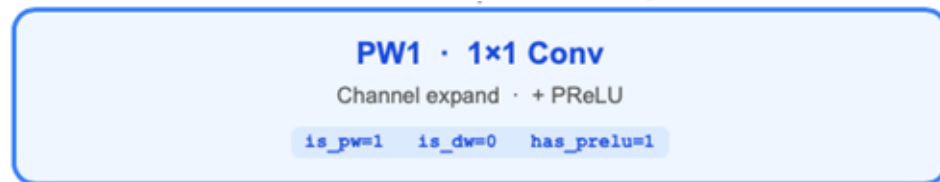
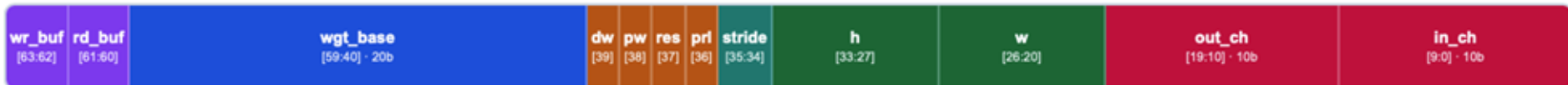
✓ HWC LAYOUT (CHANNEL-LAST) — OUR CHOICE

Memory order: all channels at $(0,0)$ → all channels at $(0,1)$ → ...



✓ To read R, G, B at position (y, x) : addresses are **contiguous**
→ Sequential read, no striding → simple address generator

Hardware - Bottleneck?



Triple-Buffer for Residual



Each layer config sets **rd_buf** and **wr_buf**

Hardware computes: `res_buf = 3 - rd_buf - wr_buf`

→ Three buffers never conflict, no stall needed

Hardware - Linear7-DW(7*6)?

△ PROBLEM: KERNEL HAS 42 TAPS, MAC ARRAY ONLY HAS 9

7×6 kernel (42 taps)

0	1	2	3	4	5
6	7	8	9	10	11
12	13	14	15	16	17
18	19	20	21	22	23
24	25	26	27	28	29
30	31	32	33	34	35
36	37	38	39	40	41

3×3 normal (9 taps)

≠

0	1	2
3	4	5
6	7	8

✓ SOLUTION: SPLIT INTO 5 GROUPS × 9, REUSE MAC ARRAY 5 TIMES

0	1	2	3	4	5	6	7	8	Group 0 · tap 0–8	→ psum ₀
9	10	11	12	13	14	15	16	17	Group 1 · tap 9–17	→ psum ₁
18	19	20	21	22	23	24	25	26	Group 2 · tap 18–26	→ psum ₂
27	28	29	30	31	32	33	34	35	Group 3 · tap 27–35	→ psum ₃
36	37	38	39	40	41	0	0	0	Group 4 · tap 36–41 + 3 zeros	→ psum ₄

Final output = psum₀ + psum₁ + psum₂ + psum₃ + psum₄

per channel × 512

- The MAC array, sliding window, and accumulator are unchanged.
 - Only the controller changes behavior:
 - Instead of 1 MAC pass → 5 MAC passes with accumulated psum.
- Zero hardware redesign needed !!

Verification

- Currently verified our top module with
 - C model golden vector in each layer
 - Latent embedding similarity among different images

RTL vs C Model - Embedding Exact Match			
Image	Match	MaxDiff	MeanAbsErr
dan_1.jpg	128/128	0	0.0000
dan_2.jpg	128/128	0	0.0000
dan_3.jpg	128/128	0	0.0000
elon_1.jpg	128/128	0	0.0000
elon_2.jpg	128/128	0	0.0000
elon_3.jpg	128/128	0	0.0000
elon_4.jpg	128/128	0	0.0000
trump_1.jpg	128/128	0	0.0000
trump_2.jpg	128/128	0	0.0000
trump_3.jpg	128/128	0	0.0000
wil_1.jpg	128/128	0	0.0000
wil_2.jpg	128/128	0	0.0000
wil_3.jpg	128/128	0	0.0000

Summary Comparison		
	C Fixed	RTL
Intra avg	0.5270	0.5270
Inter avg	0.0783	0.0783
Gap	+0.4487	+0.4487

[Verify] Comparing 50 layers ...				
Layer	Exact%	±1%	MeanErr	MaxErr
L00	100.00%	100.00%	0.000	0
L01	100.00%	100.00%	0.000	0
L02	100.00%	100.00%	0.000	0
L03	100.00%	100.00%	0.000	0
L04	100.00%	100.00%	0.000	0
L05	100.00%	100.00%	0.000	0
L06	100.00%	100.00%	0.000	0
L07	100.00%	100.00%	0.000	0
L08	100.00%	100.00%	0.000	0
L09	100.00%	100.00%	0.000	0
L10	100.00%	100.00%	0.000	0
L11	100.00%	100.00%	0.000	0
L12	100.00%	100.00%	0.000	0
L13	100.00%	100.00%	0.000	0
L14	100.00%	100.00%	0.000	0
L15	100.00%	100.00%	0.000	0
L16	100.00%	100.00%	0.000	0
L17	100.00%	100.00%	0.000	0
L18	100.00%	100.00%	0.000	0
L19	100.00%	100.00%	0.000	0
L20	100.00%	100.00%	0.000	0
L21	100.00%	100.00%	0.000	0
L22	100.00%	100.00%	0.000	0
L23	100.00%	100.00%	0.000	0
L24	100.00%	100.00%	0.000	0
L25	100.00%	100.00%	0.000	0
L26	100.00%	100.00%	0.000	0
L27	100.00%	100.00%	0.000	0
L28	100.00%	100.00%	0.000	0
L29	100.00%	100.00%	0.000	0
L30	100.00%	100.00%	0.000	0
L31	100.00%	100.00%	0.000	0
L32	100.00%	100.00%	0.000	0
L33	100.00%	100.00%	0.000	0
L34	100.00%	100.00%	0.000	0
L35	100.00%	100.00%	0.000	0
L36	100.00%	100.00%	0.000	0
L37	100.00%	100.00%	0.000	0
L38	100.00%	100.00%	0.000	0
L39	100.00%	100.00%	0.000	0
L40	100.00%	100.00%	0.000	0
L41	100.00%	100.00%	0.000	0
L42	100.00%	100.00%	0.000	0
L43	100.00%	100.00%	0.000	0
L44	100.00%	100.00%	0.000	0
L45	100.00%	100.00%	0.000	0
L46	100.00%	100.00%	0.000	0
L47	100.00%	100.00%	0.000	0
L48	100.00%	100.00%	0.000	0
L49	100.00%	100.00%	0.000	0
Status: ALL PASS				

TODO

Priority	Phase	Task
● Now	Synthesis	Run Design Compiler, check timing report
● Now	Synthesis	Fix critical path: addr_gen 2-stage pipeline + activation PReLU register
● Near	System Integration	Replace Weight ROM → SRAM + AXI-Lite write interface
● Near	System Integration	AXI-Stream DMA interface for feature map I/O
● Ongoing	Datapath Upgrade	Replace 1×1 PW datapath with Systolic Array
● Long	Backend	Innovus ASIC backend flow (floorplan → routing → GDSII)
● Near	Verification?	FPGA integration

Gantt Chart

WBS NUMBER	TASK TITLE	Reference Link	TASK OWNER	START DATE	DUE DATE	DURATION	PCT OF TASK COMPLETE	WEEK 1					WEEK 2					WEEK 3					WEEK 4					WEEK 5																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
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Deliverables

- Top-level architecture and block diagrams of the MobileFaceNet frontend accelerator
- RTL design of the main accelerator modules
- Fixed-point C reference model and bit-level functional verification against PyTorch golden output
- Synthesis results and timing analysis
- Runtime weight update support via writable weight SRAM and AXI-Lite interface
- Final report and presentation



References

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- [3] S. Chen, Y. Liu, X. Gao, and Z. Han, Mobilefacenets: Efficient cnns for accurate real-time face verification on mobile devices, 2018. arXiv: 1804 . 07573 [cs.CV]. [Online]. Available: [https : / / arxiv.org/abs/1804.07573](https://arxiv.org/abs/1804.07573).